Day-one risk coupons: test sheet

Print before the big parts; every coupon uses the real part's geometry functions and numbers. As exported, no supports, 0.4 nozzle, 0.2 layers, 4 walls. Plates: risk_plate_1.3mf (PLA, 19), risk_plate_2.3mf (PPA-CF, 26). Change one number at a time, rerun python -m tools.risk_coupon, reprint that coupon.

#	do	pass	if it fails, change (file: name)
1 Swing-arm direction splitter. Bought: O-ring 20 x 1.5 (arm hub groove), e-clip for the d8 arm pin, 2 x M3x16	O-ring into the arm hub groove. On the plate: sleeve on the centre pin, sun on its rib, arm on the sleeve above the sun with its pin down through the idler, e-clip on the pin, output on its pin, crank on top, stop finger on the column (2 x M3x16). Turn the crank 10 turns each way.	Clockwise seen from above: the arm swings the idler into the output at once and the output turns, no skipped teeth. Anticlockwise: the arm swings out to the finger within about 20 deg of crank, the idler runs free and the output stays still. No jam at engagement.	launch_aim.py: FIT (pin and bore play), arm_solid O-ring groove (drag), A() alpha_d = alpha_e + 20 (swing out), _stop_post_xy; launch_gears.py center_distance extra / spur backlash (mesh).
2 Printed contact switch. Bought: M3x10 SHCS ISO 4762 + M3 jam nut, M3 nut, 2x M3x10 self tapping (pan or SHCS) through the lever root into the pad, at most M3x11.7, 2 wires, multimeter	Nut into the pillar trap (drop of glue), lever clamped with 2 M3 self tappers, contact screw head down through the lever, set the head 0.9 mm above the nut (0.5 mm per turn), lock the jam nut. Meter on beep between screw and nut. Press the lever tip 100 times.	Beeps after about 1.2 mm of tip travel, silent when let go, tip stops on the hard stop without cracking, lever back flat, same after 100 presses.	cad/lib/switch.py contact_switch args travel, overtravel, t, b; STRAIN_LIMIT, NUT_PRESS (loose nut), SCREW_LENGTHS.
3 Dog clutch and compliant fork. Bought: 1 x M3x12 (fork stop), M3 nut (tab), 2 x M3 screws (SW_DOG prongs, optional)	Press the rail into the base. REEL face on the journal (dogs up), collar on the hex (dogs down), fork pads in the collar groove and its sleeve on the rail, fork stop on (M3). Push the fork 7 mm down with the dogs tip to tip, then turn the REEL face until they drop in. Turn the face hard both ways, then pull the fork back.	Yoke flexes when tip to tip (no jam, roofs slide off), dogs drop in within a quarter turn, under hand torque the dogs pull in, never walk out, a light pull on the fork releases them, the yoke springs back flat.	_drum_kin.py DEFAULTS dog_backcut, dog_roof, dog_deg, dog_gap, fork_k; drum.py GEOM fork_t, yoke_r, groove; fits.py SHAFT_IN_BORE (collar on the hex, face on the journal), HOLE_SLIDE (fork on the rail).
4 Face slip clutch with the flexure spring. Bought: 3 x M3x20 SHCS, 3 x M3 nut, string, luggage scale	Stack on the key stub: hub plate (counterbores down), friction disc, pressure plate (pads up), flexure (beam down, tabs on the screws), adjuster plate (traps up); 3 screws from below, nuts on top. Run the nuts down until the flexure just touches, then turn each nut 1/2 turn at a time (0.25 mm). Clamp the stub, tie a string to a disc lug (r 28.5 mm), pull along the rim with the luggage scale while it slips steadily.	Smooth steady slip, no judder; the reading repeats within 20 percent and rises with each half turn, close to the table below (kg on the scale at the lug); the nuts never bottom the flexure before the last row. Expected: 0.12 mm: 0.013 N m, 0.05 kg; 0.25 mm: 0.027 N m, 0.10 kg; 0.38 mm: 0.040 N m, 0.14 kg; 0.50 mm: 0.053 N m, 0.19 kg.	cad/lib/clutch.py: slip_clutch args flex_t, flex_gap, flex_ro, r_in, r_out, n_grooves; MU_PLA_PLA, E_PLA, SIGMA_FLEX_PLA_SUSTAINED (creep: if the torque falls overnight).
5 Bearing seats. Bought: 608ZZ, F693ZZ, 693ZZ, CSK20PP	Press each bearing in square with a flat block or a vice (tile left to right: 608, F693ZZ, 693ZZ); CSK20PP into the round boss with its outer keyway on the printed key.	Firm press, seated flat on the lip, stays in when tapped upside down, spins as freely as before; the CSK20PP key slides in without force, it still freewheels one way and locks the other.	fits.py BRG_RIB_INTERF, BRG_RIB_BORE (all seats); bearings.py CSK20_KEY_RCLEAR, CSK20_OKEY_W (key), KEY_CLEAR in fits.py. Print the fit coupon tile A if the 608 is off.
6 M3 holes. Bought: M3x4x5 heat-set insert, M3 nut, 2 x M3x10	Left to right: heat-set the insert (220 C), press the nut in, push a screw through, self-tap a screw in.	Insert flush and does not spin when the screw is tightened; nut presses in and holds; screw drops through the clearance hole; the tap hole cuts a thread and holds a hard pull without splitting.	fits.py M3_INSERT_D, M3_NUT_AF, M3_CLEAR_D, M3_TAP_D (the fit coupon tiles C, D, E bracket them).
7 Key rib and spigot. Bought: CSK20PP from coupon 5 (inner keyway)	Slide the key stub into the keyed tile and into the CSK20PP inner race (the real interface). Push the spigot into the sleeve slice and turn it.	Stub slides in by hand, key in the keyway, no visible rock, under 1 deg of play when turned; the spigot turns freely with no wobble.	fits.py KEY_CLEAR, SHAFT_IN_BORE; shafts.py KEY20_H, SHAFT_PRINT_UNDERSIZE; launch.py SPIGOT_BORE (now 12.35); drum.py GEOM spig_d.
8 Gear meshes m1.25, m1.5, m2, m2.5. Bought: none	Each pair on its plate (label = module), flat faces down; turn or rock through every tooth.	Smooth, no tight spot, a little play when rocked; m2.5: the tall tooth and the cut-down last tooth pass without catching, then the mesh lets go at the gap.	fits.py GEAR_BACKLASH, GEAR_CENTER_EXTRA (m1.25, m1.5, m2); launch_gears.py center_distance extra, spur backlash, sector_gear_2d tall_factor (m2.5); _drum_kin.py pin_x (m1.25 pinion).
9 Shear-pin torque limiter. Bought: string or a strap, luggage scale	Hub slice into the outer slice, pin through both holes (head outside). Clamp the hub lever, hang the luggage scale on the outer lever hole (r 60 mm) at right angles and pull slowly until the pin shears. Break all 3 pins. Torque = kg x 9.81 x 0.06 N m.	Clean single shear at about 8.3 kg (4.9 N m), the 3 pins within 20 percent, both slices undamaged (holes still round); a fresh pin goes in by hand.	drum.py GEOM shear_d (torque goes with d squared), hub_od (radius), shear_s; fits for the hole: drum.coupler shear hole (shear_d + 0.1) and shear_pin (shear_d - 0.04). Spec D8 / D26 limit 5 N m.